

```
import matplotlib.pyplot as plt
# Pie chart
labels = ['Python', 'Java', 'Ruby', 'C++']
sizes = [32.8, 18.2, 27, 21.9]
# colors
colors = ['#ff9999', '#66b3ff', '#99ff99', '#ffcc99']
fig1, ax1 = plt.subplots()
patches, texts, autotexts = ax1.pie(sizes, colors = colors, labels=labels, autopct='%1.1f%%', startangle=90)
for text in texts:
    text.set_color('grey')
for autotext in autotexts:
    autotext.set_color('grey')
# Equal aspect ratio ensures that pie is drawn as a circle
ax1.axis('equal')
plt.tight_layout()
plt.show()
```

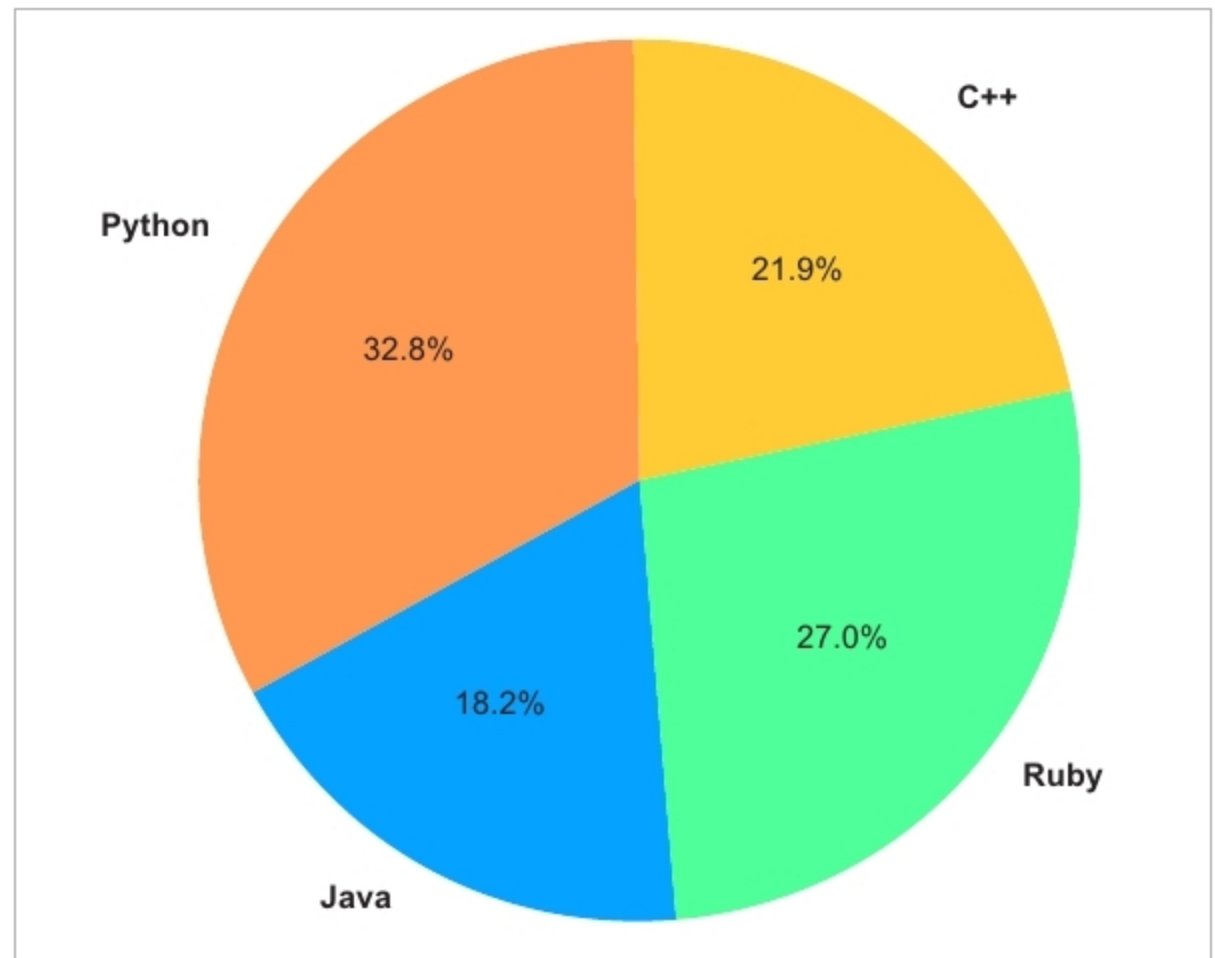


Figure 5: Plotting a pie chart using Matplotlib

6) Visualisation using a box plot

plot: A box plot is the visual representation depicting groups of numerical data through their quartiles. It is used to analyse how distributed the data is across the data sets. Box plots summarise the sample data using the 25th, 50th and 75th percentiles. These percentiles are also known as the lower, median and upper quartile.

```
import matplotlib.pyplot as plt
np.random.seed(10)
one=np.random.normal(100,10,200)
two=np.random.normal(80, 30, 200)
three=np.random.normal(90, 20, 200)
four=np.random.normal(70, 25, 200)
to_plot=[one,two,three,four]
fig=plt.figure(1,figsize=(9,6))
ax=fig.add_subplot(111)
bp=ax.boxplot(to_plot)
fig.savefig('boxplot.png',bbox_inches='tight')
```

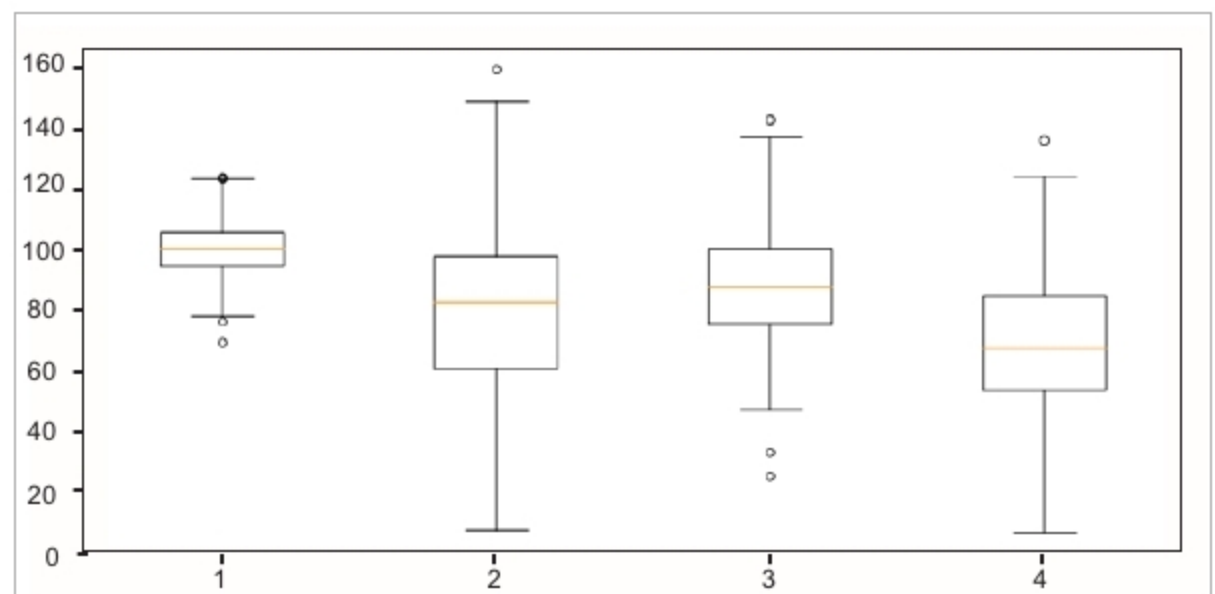


Figure 6: Plotting a box plot using Matplotlib

does not make sense until the user can look at it in a visual form, such as with charts and plots. This article will help data scientists to understand the different types of plots that are needed when visualising data in Python. **END** 🐧

References

- [1] <https://www.anaconda.com/distribution/>
- [2] <https://seaborn.pydata.org/>
- [3] <https://matplotlib.org/>
- [4] <https://plotly.com/python/>
- [5] <https://github.com/andrea-cuttone/geoplotlib>

By: Dr Aditya Bhardwaj

The author is a B.Tech, M.Tech and Ph.D in CSE. Currently, he is assistant professor at PEC (a deemed university), Chandigarh. He has experience in cloud computing and Big Data technology.